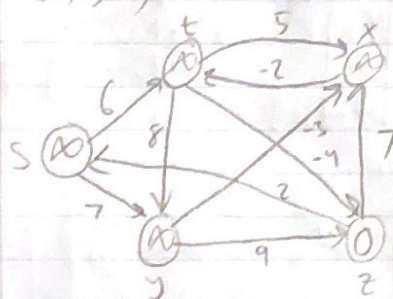


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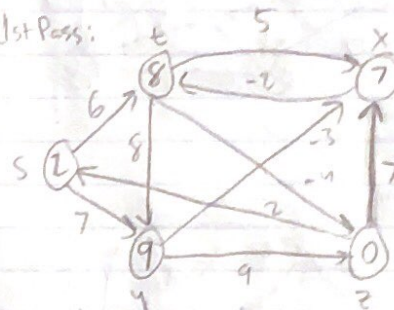
Problem Set 3

22.1-1 Run the Bellman-Ford algorithm on the directed graph of Figure 22.4, using vertex z as the source. In each pass, relax edges in the same order as in the figure, and show the d and π values after each pass. Now, change the weight of edge (z, x) to 4 and run the algorithm again, using s as the source. (Figure 22.4 says that each pass relaxes the edges in the order of: (t, x) , (t, y) , (t, z) , (x, t) , (y, x) , (y, z) , (z, x) , (z, s) , (s, t) , (s, y))



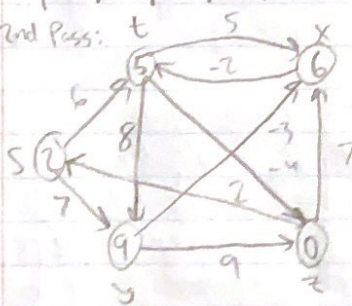
	s	t	x	y	z
d	∞	∞	∞	∞	0
π	NIL	NIL	NIL	NIL	NIL

1st Pass:



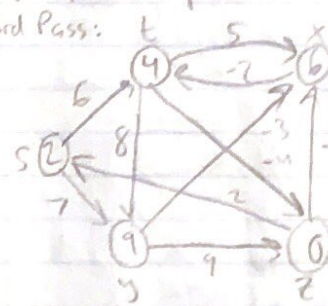
	s	t	x	y	z
d	2	8	7	9	0
π	z	s	z	s	NIL

2nd Pass:



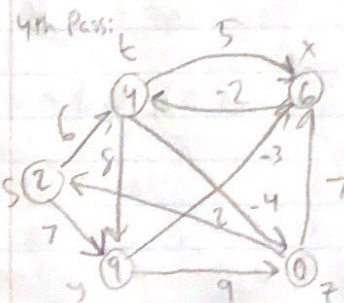
	d	π
s	2	z
t	5	x
x	6	y
y	9	s
z	0	NIL

3rd Pass:



	d	π
s	2	z
t	4	x
x	6	y
y	9	s
z	0	NIL

4th Pass:

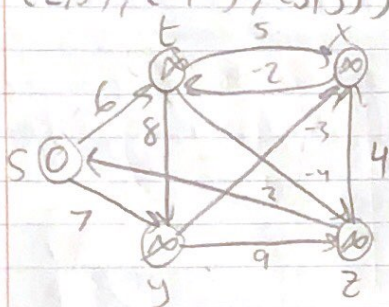


	d	π
s	2	z
t	4	x
x	6	y
y	9	s
z	0	NIL

No negative cycles, so this algorithm will return true.

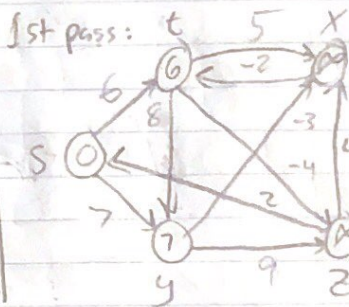
22.1-1 (continued)

Let the weight of edge (z,x) equal 4 and let's run Bellman-Ford again using "s" as the source. (Figure 22.4 says that each pass relaxes the edges in the order of: $(t,x), (t,y), (t,z), (x,t), (y,x), (y,z), (z,x), (z,s), (s,t), (s,y)$)



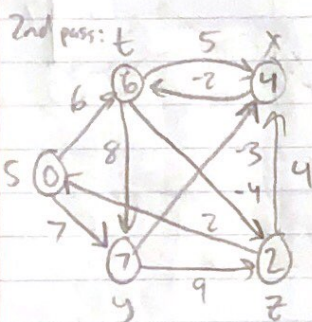
	d	π
s	0	NIL
t	∞	NIL
x	∞	NIL
y	∞	NIL
z	∞	NIL

1st pass: t



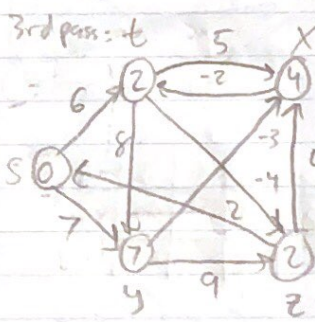
	d	π
s	0	NIL
t	6	s
x	8	NIL
y	7	s
z	2	NIL

2nd pass: t



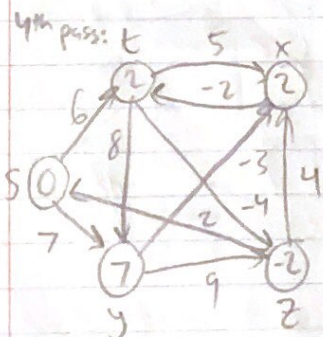
	d	π
s	0	NIL
t	6	s
x	4	y
y	7	s
z	2	t

3rd pass: t



	d	π
s	0	NIL
t	2	x
x	4	y
y	7	s
z	2	t

4th pass: t

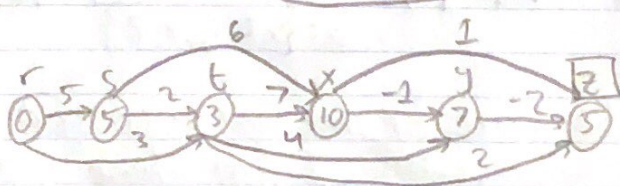
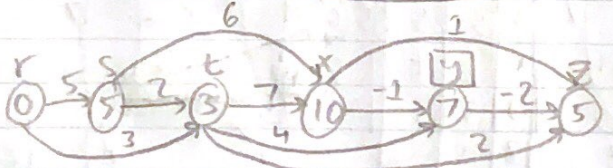
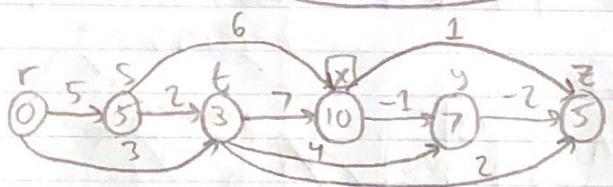
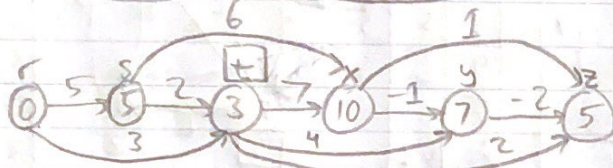
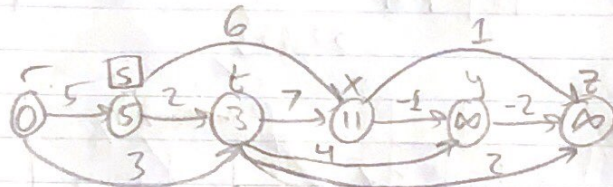
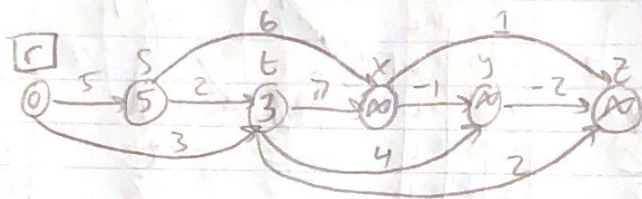
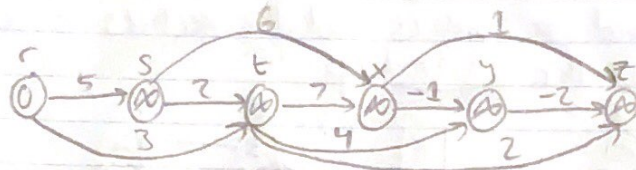


	d	π
s	0	NIL
t	2	x
x	2	z
y	7	s
z	-2	t

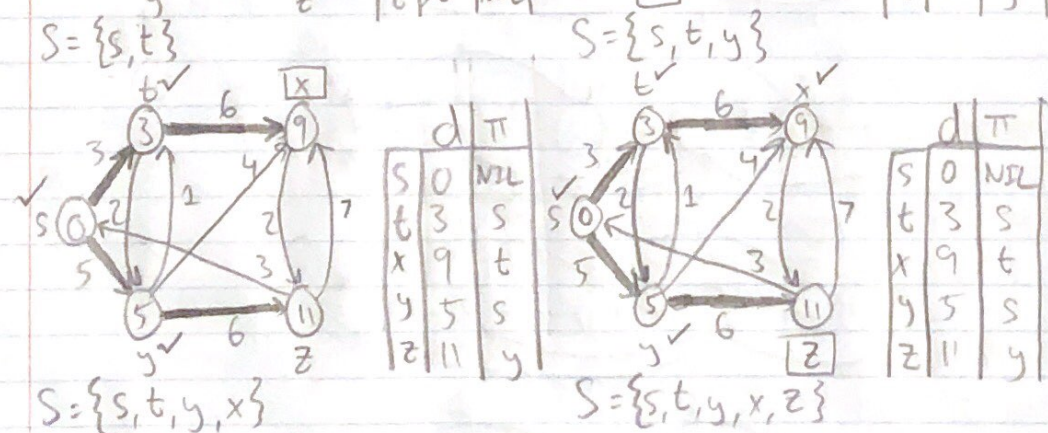
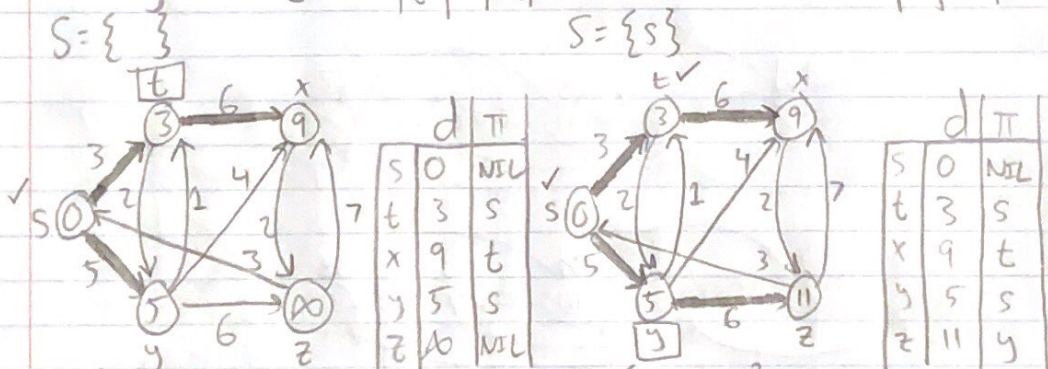
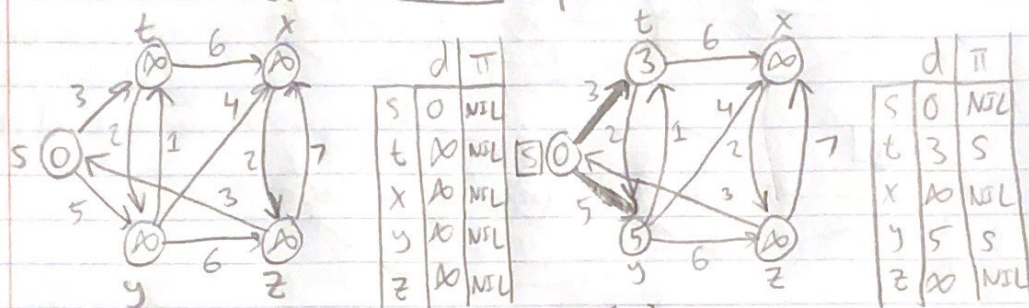
This algorithm will return false, since there is a negative weight cycle between the vertices: $\{z, x, t\}$. Since Bellman-Ford has a second for-loop to check for negative cycles, I noticed that for edge (x,t) , $t.d$ was not greater than $x.d + w(x,t)$: $t.d = 2$, $x.d = 2$, $w(x,t) = -2$, and $2 > 2 + (-2)$ is false, so this algorithm would return false.

22.2-1 Show the result of running DAG-SHORTEST-PATHS on the directed acyclic graph of Figure 27.5, using vertex "r" as the source.

★ I show the vertex that I was on by putting a box: ☐, around it



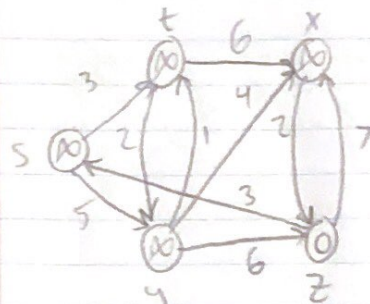
22.3-1 Run Dijkstra's algorithm on the directed graph of Figure 22.2, first using vertex "s" as the source and then using vertex "z" as the source. In the style of Figure 22.6, show the "d" and " π " values and the vertices in set S after each iteration of the while loop



★ The thick lines are for the predecessors
 ★ The "✓" is to show that I am done with that vertex
 ★ The box: $\square$, is to show which vertex I am on.

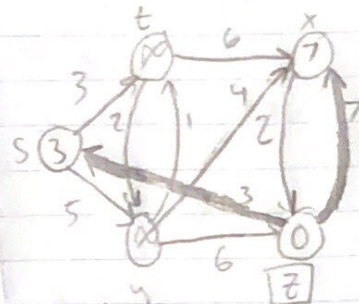
22.3-1 (continued)

Performing Dijkstra's algorithm with vertex "z" as the source.



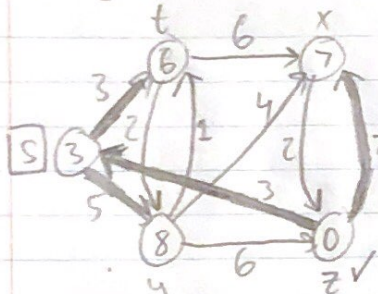
	d	π
s	∞	NIL
t	∞	NIL
x	∞	NIL
y	∞	NIL
z	0	NIL

$S = \{z\}$



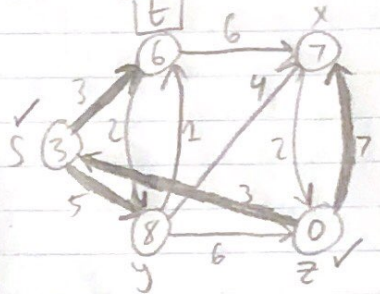
	d	π
s	3	z
t	∞	NIL
x	7	z
y	∞	NIL
z	0	NIL

$S = \{z, s\}$



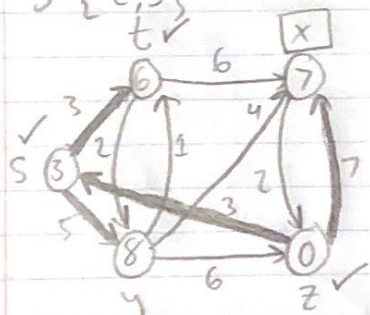
	d	π
s	3	z
t	6	s
x	7	z
y	8	s
z	0	NIL

$S = \{z, s, t\}$



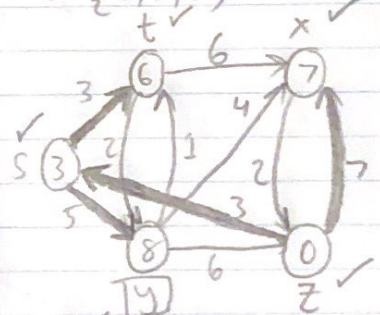
	d	π
s	3	z
t	6	s
x	7	z
y	8	s
z	0	NIL

$S = \{z, s, t, x\}$



	d	π
s	3	z
t	6	s
x	7	z
y	8	s
z	0	NIL

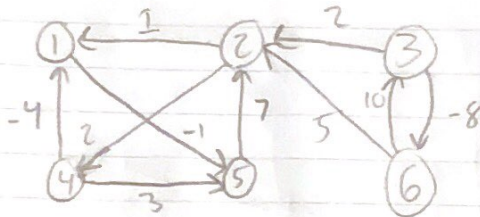
$S = \{z, s, t, x, y\}$



	d	π
s	3	z
t	6	s
x	7	z
y	8	s
z	0	NIL

$S = \{z, s, t, x, y\}$

23.2-1 Run the Floyd-Warshall algorithm on the weighted, directed graph of Figure 23.2. Show the $D^{(k)}$ that results for each iteration of the outer loop.



$$W = \begin{bmatrix} 0 & \infty & \infty & \infty & -1 & \infty \\ 1 & 0 & \infty & 2 & \infty & \infty \\ \infty & 2 & 0 & \infty & \infty & -8 \\ -4 & \infty & \infty & 0 & 3 & \infty \\ \infty & 7 & \infty & \infty & 0 & \infty \\ \infty & 5 & 10 & \infty & \infty & 0 \end{bmatrix} = D^{(0)}$$

$$D^{(1)} = \begin{bmatrix} 0 & \infty & \infty & \infty & -1 & \infty \\ 1 & 0 & \infty & 2 & 0 & \infty \\ \infty & 2 & 0 & \infty & \infty & -8 \\ -4 & \infty & \infty & 0 & -5 & \infty \\ \infty & 7 & \infty & \infty & 0 & \infty \\ \infty & 5 & 10 & \infty & \infty & 0 \end{bmatrix}$$

$$D^{(2)} = \begin{bmatrix} 0 & \infty & \infty & \infty & -1 & \infty \\ 1 & 0 & \infty & 2 & 0 & \infty \\ 3 & 2 & 0 & 4 & 2 & -8 \\ -4 & \infty & \infty & 0 & -5 & \infty \\ 8 & 7 & \infty & 9 & 0 & \infty \\ 6 & 5 & 10 & 7 & 5 & 0 \end{bmatrix}$$

$$D^{(3)} = \begin{bmatrix} 0 & \infty & \infty & \infty & -1 & \infty \\ 1 & 0 & \infty & 2 & 0 & \infty \\ 3 & 2 & 0 & 4 & 2 & -8 \\ -4 & \infty & \infty & 0 & -5 & \infty \\ 8 & 7 & \infty & 9 & 0 & \infty \\ 6 & 5 & 10 & 7 & 5 & 0 \end{bmatrix}$$

$$D^{(4)} = \begin{bmatrix} 0 & \infty & \infty & \infty & -1 & \infty \\ -2 & 0 & \infty & 2 & -3 & \infty \\ 0 & 2 & 0 & 4 & -1 & -8 \\ -4 & \infty & \infty & 0 & -5 & \infty \\ 5 & 7 & \infty & 9 & 0 & \infty \\ 3 & 5 & 10 & 7 & 2 & 0 \end{bmatrix}$$

$$D^{(5)} = \begin{bmatrix} 0 & 6 & \infty & 8 & -1 & \infty \\ -2 & 0 & \infty & 2 & -3 & \infty \\ 0 & 2 & 0 & 4 & -1 & -8 \\ -4 & 2 & \infty & 0 & -5 & \infty \\ 5 & 7 & \infty & 9 & 0 & \infty \\ 3 & 5 & 10 & 7 & 2 & 0 \end{bmatrix}$$

$$D^{(6)} = \begin{bmatrix} 0 & 6 & \infty & 8 & -1 & \infty \\ -2 & 0 & \infty & 2 & -3 & \infty \\ -5 & 3 & 0 & -1 & -6 & -8 \\ -4 & 2 & \infty & 0 & -5 & \infty \\ 5 & 7 & \infty & 9 & 0 & \infty \\ 3 & 5 & 10 & 7 & 2 & 0 \end{bmatrix}$$